YUZE LI

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EDUCATION

College of Optical Science and Engineering, Zhejiang University — **Zhejiang, China**
B.S., Optical Engineering — *Expected June 2026*

- GPA: 4.09/4.3 (3.94/4.0)
- Rank: **3/97**

SELECTED RESEARCH EXPERIENCE

Undergraduate Research Intern — Yale University
Advisor: Prof. Hong Tang (Dept. of Electrical & Computer Engineering) — Jul. 2025 - Now

- Fabrication of periodically poled nonlinear crystal bulks.
- Received a series of on-chip calibration and measurement.

Fluorescence Image Restoration Based on Neural Networks — Zhejiang University
Advisor: Prof. Huafeng Liu (Dept. of Optical Science and Engineering) — Nov. 2024 - Now

- The primary objective of this project is to leverage the robustness of Implicit Neural Representations (INR) to Gaussian and Poisson noise as a novel approach for enhancing medical images. By use INR as a prior in the whole network, we managed to denoise medical fluorescence images heavily affected by several types of noise. The improved model has already achieved promising results.

Mode field diameter controlling with laser direct writing method — Zhejiang University
Advisor: Prof. Jianrong Qiu (Dept. of Optical Science and Engineering) — Sept. 2024 – Jun. 2025

- The main work I did in this project including the fabrication (directly written in glass bulks by femtosecond laser), polishing, calibration and measurement of the in-bulk waveguides. I have experimentally verified that the mode field diameter of the waveguide can indeed exhibit regular variations by using the bulks I made.

Micro-nano fiber tapering system — Zhejiang University
Advisor: Dr. Yaoguang Ma (Dept. of Optical Science and Engineering) — Nov. 2023 - Jun. 2024

- The main focus of the experiment was on fiber morphology fitting, heating zone calibration, and the fabrication of long-waist MNFs (micro-nano fibers) with a diameter at the micrometer scale using a vertical electrically heated MNF tapering system. I also developed a custom Python algorithm to automate the measurement of optical fiber diameters under a microscope, achieving a measurement accuracy of over 95%.

PUBLICATIONS

Yuze Li, H. Liu et al. *Self-Supervised Denoising for Fluorescence Microscopy Using Image Representations as a Prior.* (Article under Submission)

Yuze Li, J. Qiu et al. *Tapered three-dimensional waveguide with controllable mode field diameter by composite laser direct writing method.* (Poster presented at 6th Ultrafast Laser Precision Processing Technology and Application Symposium)

PATENT

Yue Li, **Yuze Li (primary contributor)** et al. *A self-supervised denoising method for two-dimensional fluorescence microscopy images based on image representation.* Application pending.

HONORS

- Second Prize in the 13th National College Student Optoelectronic Design Competition, ***National-level.*** — 2025
- External Scholarship, Yongping Scholarship (Student representative in Duan's Homecoming Meeting). — 2023-2024
- First Prize in the National College Students' Mathematical Contest, ***Province-level.*** — 2023
- University-level Second Class Scholarship. — 2022-2023
- Third Prize in the Zhejiang Province Physics Innovation Competition. — 2023

PUBLIC SERVICE

- Engaged as a team leader in public legal education activities organized by the Procuratorate. — Jul. 2024 - Aug. 2024
- Volunteer service at campus post stations. — Nov. 2022-May. 2023
- Volunteer service at several conferences. — Nov. 2022

SKILLS

Programming & Software: C, Python, MATLAB, OriginPro, Fiji, ImageJ.

Experimental Skills: Measurement and calibration of ring resonators, waveguides and fibers. Platform controlling. Operation of Confocal Microscope and SEM.

Language: TOEFL ***100*** out of 120. (Reading 25, Listening 30, Speaking 22, Writing 23)